

AX

AGENTROPIX-SIFT · DFIR

FULL PROFILE · REAL-DATA RUN (2026-06-06)

AMF Windows sample001 — Memory Injection Triage

Single-host Windows XP physical memory image · DFIR triage · process injection (T1055)

REPORT ID	22560e04e9b123a0d3e09776cb5cc4cca2282c1108e009e03ed12e691612a33a
CASE ID	AMF-WIN-SAMPLE001
CASE NAME	AMF Windows sample001 (Art of Memory Forensics)
EXAMINER	victor.galvan
PROFILE	full
INCIDENT TYPE	dfir
SEVERITY	medium
SNAPSHOT	2026-06-07T03:52:18.683321+00:00
GENERATED	2026-06-07T04:21:03Z
MCP HOST	

SEALED · report 22560e04e9b123a0... · 6 approved findings · first-finding seal hmac-sha256:fd05a9d4c2721e5a693b03b55d06ceea7cbe8429056fa93338800baeb8ed9cee · examiner victor.galvan · approved 2026-06-06T23:17:43Z (NOTE: F-AMF-S001-001 approval was Playwright-automated demo, not human HMAC sign-off)

01 · EXECUTIVE DASHBOARD

Executive Summary

Single-host triage of a 511 MiB raw Windows XP-era physical memory image (sample001.bin, the Art-of-Memory-Forensics teaching sample). Scope is RAM only — there is no on-disk corpus and no live host. Agentropix recovered 21 running processes, 229 services (118 running), a coherent two-root PPID forest (0 orphans, 0 LOLBin flags) and all 21 command lines, then ran malfind over the image and surfaced 15 executable VAD regions — 11 PAGE_EXECUTE_READWRITE (RWX) and 4 PAGE_EXECUTE_READ (RX) — concentrated in winlogon.exe (PID 628). Six findings were promoted and rolled up into this sealed full-profile report. No network endpoints and no IOCs were recoverable from this image.

Headline: 9 RWX (PAGE_EXECUTE_READWRITE) injected/unpacked-code regions in winlogon.exe (PID 628) — the dominant, canonical process-injection signal (MITRE T1055) in this image — alongside single RWX regions in csrss.exe and msimn.exe. Severity is capped at **medium**: this is a known teaching sample and the highest-confidence signal is memory-resident with no corroborating network or on-disk artefact.

6

APPROVED FINDINGS

medium×3 · low×3

1

HOSTS IN SCOPE

sample001 (memory image)

0

IOCS CATALOGUED

none recoverable from RAM

~3 h 54 m

DWELL / WINDOW

boot → acquisition (RAM
clock)

2

ATT&CK TECHNIQUES

T1055 · T1057

N/A

INITIAL ACCESS

no vector in RAM-only
scope

Hosts / evidence roster



sample001

Windows XP host · physical RAM
image · 511 MiB · injected-code
present

sample001.bin

Evidence · SHA-256 0324...d561c ·
536,330,240 bytes

02 · RISK ASSESSMENT

Risk Matrix

Six approved findings scored qualitatively. The two medium-confidence RWX concentrations in winlogon.exe (F1, F2) sit at impact 3 × likelihood 4 — credible code injection in a core logon process — but are bounded by the teaching-sample context and absence of corroborating network/disk evidence. The RX-only and process-inventory findings (F3, F5, F6) fall in the low band. *Scoring is qualitative (operator-derived from approved-finding severity + technique), not a tool-computed number.*

5

4

3

2

1

F4

F1,F2

F3,F5

F6

1

2

3

4

5

X-axis: Likelihood (1-5) · Y-axis: Impact (1-5) · Risk = Impact × Likelihood

REF	RISK	IMPACT	LIKELIHOOD	SCORE	SEVERITY
F1	malfind recovered 15 executable VAD hits (1) PAGE_EXECUTE _READWRITE / ...	3	4	12	MEDIUM
F2	9 RWX (PAGE_EXECUTE _READWRITE) malfind regions in winlogon.exe (PID 62...	3	4	12	MEDIUM
F3	2 PAGE_EXECUTE _READ (RX, not RWX) malfind regions in lsass.exe (PID 69...	2	2	4	LOW
F4	1 RWX (PAGE_EXECUTE _READWRITE) malfind region in csrss.exe (PID 604)...	3	3	9	MEDIUM
F5	1 RWX (PAGE_EXECUTE _READWRITE) malfind region in msimn.exe (PID 1984);...	2	2	4	LOW
F6	Process inventory recovered: 21 running processes, coherent PPID fores...	1	2	2	LOW

Key Findings

MEDIUM

F-AMF-S001-001

malfind recovered 15 executable VAD hits (11 PAGE_EXECUTE_READWRITE / RWX, 4 PAGE_EXECUTE_READ / RX) across 5 processes; RWX concentrated in winlogon.exe (PID 628, x9)

get_malfind (volatility3.windows.malfind.Malfind) reported 15 executable VAD hits: 11 RWX (PAGE_EXECUTE_READWRITE) and 4 RX (PAGE_EXECUTE_READ). RWX by process: winlogon.exe PID 628 x9, csrss.exe PID 604 x1, msimn.exe PID 1984 x1. RX (not RWX): winlogon.exe PID 628 x1 (0x580000), lsass.exe PID 692 x2, msmsgs.exe PID 548 x1. RWX is concentrated in winlogon.exe, the canonical AMF process-injection teaching signal.

MITRE T1055 · confidence 0.7 · host sample001

MEDIUM

F-AMF-S001-002

9 RWX (PAGE_EXECUTE_READWRITE) malfind regions in winlogon.exe (PID 628, ppid 356) — dominant injected/unpacked-code concentration

get_malfind reported 10 executable VAD hits in winlogon.exe (PID 628, ppid 356 smss.exe), of which 9 are RWX (PAGE_EXECUTE_READWRITE, vad_tag VadS) and 1 is RX (PAGE_EXECUTE_READ at 0x580000, vad_tag Vad). The 9 RWX regions are the dominant injected/unpacked-code concentration in this image and the canonical AMF process-injection teaching signal.

MITRE T1055 · confidence 0.7 · host sample001

LOW

F-AMF-S001-003

2 PAGE_EXECUTE_READ (RX, not RWX) malfind regions in lsass.exe (PID 692) — read-execute, no write permission

get_malfind reported 2 executable VAD regions in lsass.exe (PID 692, ppid 628 winlogon.exe), both PAGE_EXECUTE_READ (RX), vad_tag Vad — NOT PAGE_EXECUTE_READWRITE. An earlier draft mislabeled these as RWX injection into the credential process; the authoritative malfind output (step6_malfind_300s.json) shows protection PAGE_EXECUTE_READ for both, so these are read-execute mapped regions with no write permission and are not RWX injection. Lower confidence accordingly.

MITRE T1055 · confidence 0.5 · host sample001

MEDIUM

F-AMF-S001-004

1 RWX (PAGE_EXECUTE_READWRITE) malfind region in csrss.exe (PID 604)

get_malfind reported 1 RWX VAD region in csrss.exe (PID 604, ppid 356 smss.exe) at 0x7f6f0000, vad_tag Vad, PAGE_EXECUTE_READWRITE.

MITRE T1055 · confidence 0.65 · host sample001

LOW

F-AMF-S001-005

1 RWX (PAGE_EXECUTE_READWRITE) malfind region in msimn.exe (PID 1984); msmsgs.exe (PID 548) region is PAGE_EXECUTE_READ (RX, not RWX)

get_malfind reported 1 RWX VAD region in msimn.exe (Outlook Express, PID 1984) at 0x1eb0000, vad_tag VadS, PAGE_EXECUTE_READWRITE. The msmsgs.exe (Windows Messenger, PID 548) region at 0x520000 is PAGE_EXECUTE_READ (RX), vad_tag Vad — NOT RWX. An earlier draft labeled both as RWX; the authoritative malfind output shows only msimn.exe is RWX. RWX in these GUI apps is commonly benign JIT/packing on XP-era images, hence low confidence.

MITRE T1055 · confidence 0.5 · host sample001

LOW

F-AMF-S001-006

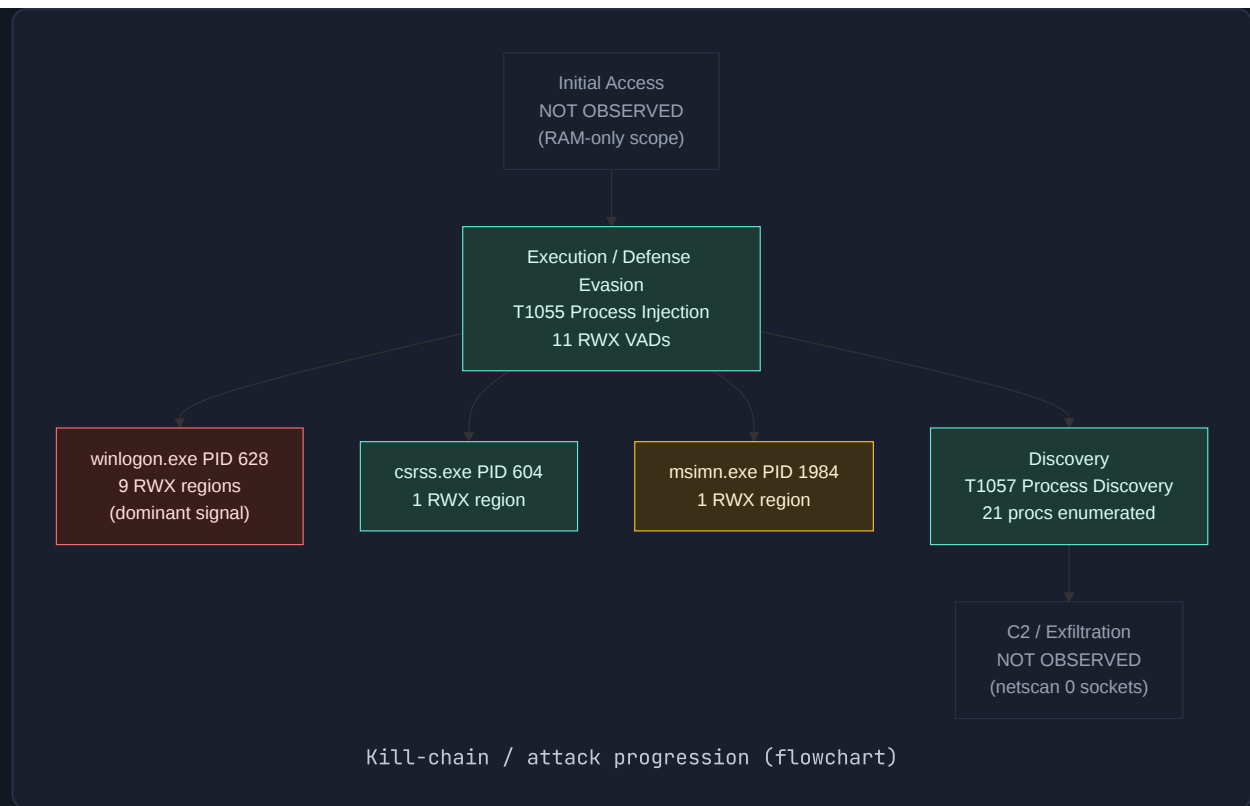
Process inventory recovered: 21 running processes, coherent PPID forest (2 roots, 0 orphans, 0 LOLBin flags)

get_pslist returned 21 processes (System pid4, smss 356, csrss 604, winlogon 628, services 680, lsass 692, svchost 852...); build_process_tree: process_count 21, root_count 2 (System, smss.exe), orphan_count 0, suspicious_count 0. cmdline recovered all 21 command lines consistent with pslist. get_netscan recovered 0 sockets (honest empty real result on this XP-era image). Baseline process-discovery context for the injection findings.

MITRE T1057 · confidence 0.6 · host sample001

Attack Chain & MITRE ATT&CK

This is a memory-only artefact set: no initial-access vector, lateral movement or exfiltration channel is reconstructable from RAM alone. What the image *does* show is the Defense Evasion / execution stage — process injection (T1055): 11 RWX VADs concentrated in winlogon.exe, the canonical injection target. The process-discovery finding maps to T1057. The chain below shows only the evidence-grounded stages; greyed stages are NOT observed in this RAM-only scope.



MITRE ATT&CK techniques observed

DEFENSE EVASION • PRIVILEGE ESCALATION

T1055

Process Injection

11 PAGE_EXECUTE_READWRITE (RWX) VAD regions across winlogon.exe (×9), csrss.exe (×1) and msimn.exe (×1); 4 PAGE_EXECUTE_READ (RX) regions in winlogon.exe, lsass.exe and msmsgs.exe. Findings F-AMF-S001-001..005, get_malfind.

DISCOVERY

T1057

Process Discovery

21 running processes enumerated via get_pslst; coherent two-root PPID forest (System, smss.exe) confirmed by build_process_tree, 0 orphans / 0 LOLBin flags. Finding F-AMF-S001-006.

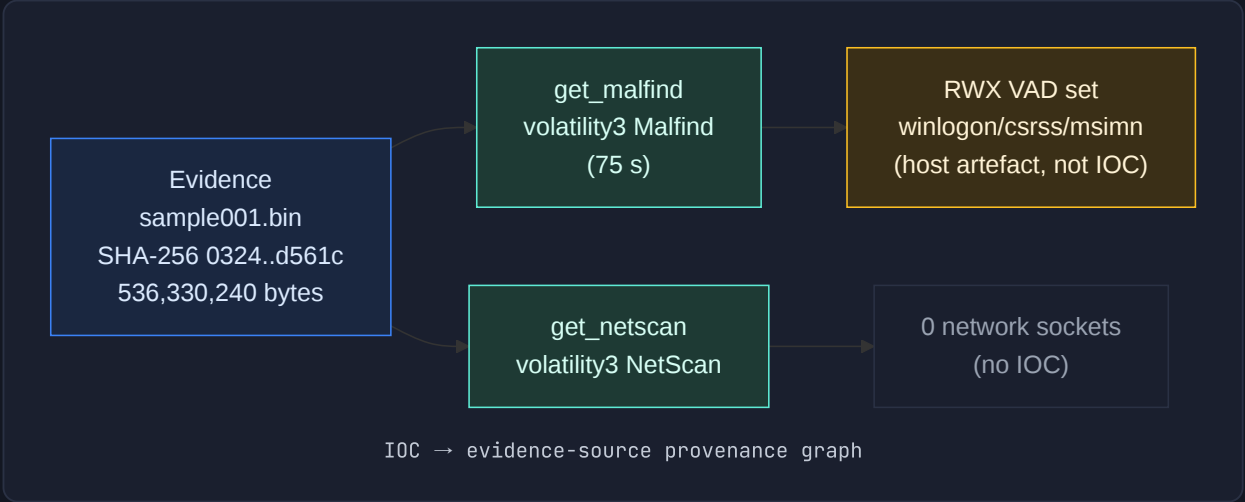
IOC Catalogue

No indicators of compromise were recoverable from this image. get_netscan returned 0 sockets (an honest empty real result on this XP-era image), no malicious file hashes or paths were extracted, and no identity/account indicators surfaced. The injected-code findings are memory-resident VAD artefacts (base addresses + protections), not network/file IOCs — they are catalogued as host artefacts (§6) rather than IOCs.

TYPE	VALUE	ROLE	CONFIDENCE	MITRE	SOURCE
No network/file/identity IOCs surfaced in this memory image — see §6 for the memory-resident RWX VAD artefacts.					

Negative space (surveyed, not observed): IPv4/IPv6 endpoints — none (netscan 0 sockets); domains/URLs — none extracted; file hashes — only the evidence-image SHA-256 (custody anchor), no malware hashes; user accounts / credentials — none catalogued. The absence of network IOCs is itself a finding: the injection signal is purely memory-resident.

IOC provenance



Host Artefacts

Single Windows XP memory image. On-disk artefacts are not applicable (RAM-only scope). The process table below is the full 21-process inventory from `get_pslist` (boot anchor 2012-11-26 22:03:28 UTC); the RWX/RX VAD column flags the malfind-recovered injected-code regions per process. 229 services were recovered (118 running) — a representative running sample follows. Note `mdd.exe` (PID 244) and `wc.exe` (PID 364) are the memory-acquisition tooling captured in-image.

sample001 — process inventory (21 processes)

PID	PPID	IMAGE	STARTED (UTC)	THREADS	WOW64	MALFIND VADS	SOURCE
4	0	System	—	51 thr	False		get_pslist
356	4	smss.exe	2012-11-26 22:03:28	3 thr	False		get_pslist
684	356	csrss.exe	2012-11-26 22:03:29	12 thr	False	RWXx1	get_pslist

PID	PPID	IMAGE	STARTED (UTC)	THREADS	WOW64	MALFIND VADS	SOURCE
628	356	winlogon.exe	2012-11-26 22:03:29	18 thr	False	RWX x9, RX x1	get_pslist
688	628	services.exe	2012-11-26 22:03:30	15 thr	False		get_pslist
692	628	lsass.exe	2012-11-26 22:03:30	22 thr	False	RX x2	get_pslist
852	688	svchost.exe	2012-11-26 22:03:31	14 thr	False		get_pslist
940	688	svchost.exe	2012-11-26 22:03:31	9 thr	False		get_pslist
1024	688	svchost.exe	2012-11-26 22:03:32	76 thr	False		get_pslist
1068	688	svchost.exe	2012-11-26 22:03:32	5 thr	False		get_pslist
1116	688	svchost.exe	2012-11-26 22:03:33	14 thr	False		get_pslist
1348	688	spoolsv.exe	2012-11-26 22:03:34	10 thr	False		get_pslist
1888	688	alg.exe	2012-11-26 22:03:35	6 thr	False		get_pslist
284	244	explorer.exe	2012-11-26 22:03:58	9 thr	False		get_pslist
548	284	msmsgs.exe	2012-11-26 22:04:03	3 thr	False	RX x1	get_pslist
556	284	ctfmon.exe	2012-11-26 22:04:03	1 thr	False		get_pslist
1628	1024	wuauclt.exe	2012-11-26 22:04:43	3 thr	False		get_pslist
1984	284	msimn.exe	2012-11-26 22:00	7 thr	False	RWX x1	get_pslist

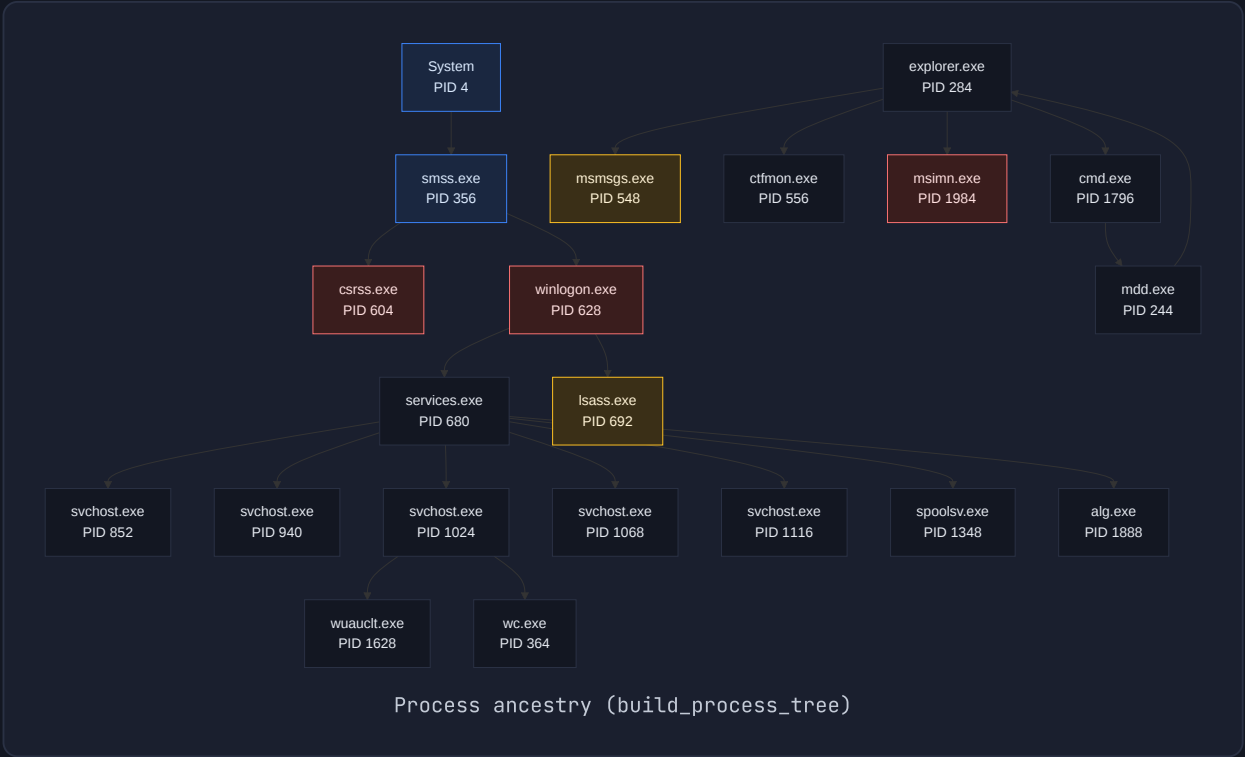
PID	PPID	IMAGE	STARTED (UTC)	THREADS	WOW64	MALFIND VADS	SOURCE
			6:33				
364	1824	wc.exe	2012-11-27 01:30:00	1 thr	False		get_pslist
1796	284	cmd.exe	2012-11-27 01:56:21	1 thr	False		get_pslist
244	1796	mdd.exe	2012-11-27 01:57:28	1 thr	False		get_pslist

Services (sample of 229 — 118 running)

SERVICE	IMAGE PATH	STATE	CLASS	SOURCE
Abiosdsk	N/A	SERVICE_STOPPED	SERVICE_KERNEL_DRIVER	get_svcs
abp480n5	N/A	SERVICE_STOPPED	SERVICE_KERNEL_DRIVER	get_svcs
ACPI	\\Driver\\ACPI	SERVICE_RUNNING	SERVICE_KERNEL_DRIVER	get_svcs
ACPIEC	N/A	SERVICE_STOPPED	SERVICE_KERNEL_DRIVER	get_svcs
adpu160m	N/A	SERVICE_STOPPED	SERVICE_KERNEL_DRIVER	get_svcs
AFD	\\Driver\\AFD	SERVICE_RUNNING	SERVICE_KERNEL_DRIVER	get_svcs
agp440	\\Driver\\agp440	SERVICE_RUNNING	SERVICE_KERNEL_DRIVER	get_svcs
Aha154x	N/A	SERVICE_STOPPED	SERVICE_KERNEL_DRIVER	get_svcs
aic78u2	N/A	SERVICE_STOPPED	SERVICE_KERNEL_DRIVER	get_svcs
aic78xx	N/A	SERVICE_STOPPED	SERVICE_KERNEL_DRIVER	get_svcs
Alert	N/A	SERVICE_STOPPED	SERVICE_WIN32_SHARE_PROCESS	get_svcs
ALG	C:\\WINDOWS\\System32\\alg.exe	SERVICE_RUNNING	SERVICE_WIN32_OWN_PROCESS	get_svcs

Full service set: 229 total — 131 kernel drivers, 63 shared-process Win32, 14 own-process Win32, 14 filesystem drivers, 7 interactive. 118 in SERVICE_RUNNING state.

Process tree



Network Artefacts

get_netscan (volatility3 NetScan) recovered 0 sockets from this image — an honest, empty real result. No ESTABLISHED or LISTENING endpoints, no remote IPs, no C2 channel are present in the capture. This is consistent with the XP-era image and bounds the attack-chain reconstruction to the host's memory-resident injection evidence.

PROCESS (PID)	LOCAL	REMOTE	STATE	PURPOSE	SOURCE
get_netscan returned 0 sockets — no network connections of interest in this capture.					

No beacon, pivot or lateral-movement evidence is reconstructable: the network layer is empty in this memory image.

Detailed Findings & Evidence

MEDIUM

F-AMF-S001-001

malfind recovered 15 executable VAD hits (11 PAGE_EXECUTE_READWRITE / RWX, 4 PAGE_EXECUTE_READ / RX) across 5 processes; RWX concentrated in winlogon.exe (PID 628, x9)

get_malfind (volatility3.windows.malfind.Malfind) reported 15 executable VAD hits: 11 RWX (PAGE_EXECUTE_READWRITE) and 4 RX (PAGE_EXECUTE_READ). RWX by process: winlogon.exe PID 628 x9, csrss.exe PID 604 x1, msimn.exe PID 1984 x1. RX (not RWX): winlogon.exe PID 628 x1 (0x580000), lsass.exe PID 692 x2, msmsgs.exe PID 548 x1. RWX is concentrated in winlogon.exe, the canonical AMF process-injection teaching signal.

MITRE T1055 · confidence 0.7 · host sample001

EVIDENCE REFS

- malfind RWX winlogon.exe PID 628 x9
(0x42e20000,0x22f40000,0x24c00000,0x248a0000,0x40a60000,0x2b7f0000,0x43400000,0x57500000,0x548a0000)
vad_tag VadS PAGE_EXECUTE_READWRITE
- malfind RWX csrss.exe PID 604 0x7f6f0000 vad_tag Vad PAGE_EXECUTE_READWRITE
- malfind RWX msimn.exe PID 1984 0x1eb0000 vad_tag VadS PAGE_EXECUTE_READWRITE
- malfind RX winlogon.exe PID 628 0x580000 vad_tag Vad PAGE_EXECUTE_READ
- malfind RX lsass.exe PID 692 0x280000,0x7f6f0000 vad_tag Vad PAGE_EXECUTE_READ
- malfind RX msmsgs.exe PID 548 0x520000 vad_tag Vad PAGE_EXECUTE_READ
- plugin volatility3.windows.malfind.Malfind hit_count=15

SOURCE ARTIFACT /cases/AMF_MemorySamples/windows/sample001.bin · SHA-256 03242077eb3364fb...316d561c

SEAL hmac-sha256:fd05a9d4c2721e5a693b03b55d06ccea7cbe8429056fa93338800baeb8ed9cee · provenance MCP · run 2f547400

MEDIUM

F-AMF-S001-002

9 RWX (PAGE_EXECUTE_READWRITE) malfind regions in winlogon.exe (PID 628, ppid 356) — dominant injected/unpacked-code concentration

get_malfind reported 10 executable VAD hits in winlogon.exe (PID 628, ppid 356 smss.exe), of which 9 are RWX (PAGE_EXECUTE_READWRITE, vad_tag VadS) and 1 is RX (PAGE_EXECUTE_READ at 0x580000, vad_tag Vad). The 9 RWX regions are the dominant injected/unpacked-code concentration in this image and the canonical AMF process-injection teaching signal.

MITRE T1055 · confidence 0.7 · host sample001

EVIDENCE REFS

- winlogon.exe PID 628 RWX VadS:
0x42e20000,0x22f40000,0x24c00000,0x248a0000,0x40a60000,0x2b7f0000,0x43400000,0x57500000,0x548a0000
(9 x PAGE_EXECUTE_READWRITE)
- winlogon.exe PID 628 RX Vad: 0x580000 (PAGE_EXECUTE_READ, not RWX)
- pslist: winlogon.exe pid 628 ppid 356
- plugin volatility3.windows.malfind.Malfind

SOURCE ARTIFACT /cases/AMF_MemorySamples/windows/sample001.bin · SHA-256 03242077eb3364fb...316d561c

SEAL hmac-sha256:40eaebdbc34767672073b7039fb37e78fc377bb70ced26b0cefe276c4ee5ab6b · provenance MCP · run c4f05906

LOW

F-AMF-S001-003

2 PAGE_EXECUTE_READ (RX, not RWX) malfind regions in lsass.exe (PID 692) — read-execute, no write permission

get_malfind reported 2 executable VAD regions in lsass.exe (PID 692, ppid 628 winlogon.exe), both PAGE_EXECUTE_READ (RX), vad_tag Vad — NOT PAGE_EXECUTE_READWRITE. An earlier draft mislabeled these as RWX injection into the credential process; the authoritative malfind output (step6_malfind_300s.json) shows protection PAGE_EXECUTE_READ for both, so these are read-execute mapped regions with no write permission and are not RWX injection. Lower confidence accordingly.

MITRE T1055 · confidence 0.5 · host sample001

EVIDENCE REFS

- lsass.exe PID 692 0x280000 vad_tag Vad PAGE_EXECUTE_READ (RX)
- lsass.exe PID 692 0x7f6f0000 vad_tag Vad PAGE_EXECUTE_READ (RX)
- pslist: lsass.exe pid 692 ppid 628
- plugin volatility3.windows.malfind.Malfind

SOURCE ARTIFACT /cases/AMF_MemorySamples/windows/sample001.bin · SHA-256 03242077eb3364fb...316d561c

SEAL hmac-sha256:9167665ab26acd1176ca7c36baa8f9fb4ccb91da35568ebabb6c672208b92276 · provenance MCP · run ae41f4b0

MEDIUM

F-AMF-S001-004

1 RWX (PAGE_EXECUTE_READWRITE) malfind region in csrss.exe (PID 604)

get_malfind reported 1 RWX VAD region in csrss.exe (PID 604, ppid 356 smss.exe) at 0x7f6f0000, vad_tag Vad, PAGE_EXECUTE_READWRITE.

MITRE T1055 · confidence 0.65 · host sample001

EVIDENCE REFS

- csrss.exe PID 604 0x7f6f0000 vad_tag Vad PAGE_EXECUTE_READWRITE (RWX)
- pslist: csrss.exe pid 604 ppid 356
- plugin volatility3.windows.malfind.Malfind

SOURCE ARTIFACT /cases/AMF_MemorySamples/windows/sample001.bin · SHA-256 03242077eb3364fb...316d561c

SEAL hmac-sha256:2b42b3df08c04dce9ea9ac22696bd6cdf90d7415eb51238b2aab6b517469cab1 · provenance MCP · run c70c95b5

LOW

F-AMF-S001-005

1 RWX (PAGE_EXECUTE_READWRITE) malfind region in msimn.exe (PID 1984); msmsgs.exe (PID 548) region is PAGE_EXECUTE_READ (RX, not RWX)

get_malfind reported 1 RWX VAD region in msimn.exe (Outlook Express, PID 1984) at 0x1eb0000, vad_tag VadS, PAGE_EXECUTE_READWRITE. The msmsgs.exe (Windows Messenger, PID 548) region at 0x520000 is PAGE_EXECUTE_READ (RX), vad_tag Vad — NOT RWX. An earlier draft labeled both as RWX; the authoritative malfind output shows only msimn.exe is RWX. RWX in these GUI apps is commonly benign JIT/packing on XP-era images, hence low confidence.

MITRE T1055 · confidence 0.5 · host sample001

EVIDENCE REFS

- msimn.exe PID 1984 0x1eb0000 vad_tag VadS PAGE_EXECUTE_READWRITE (RWX)
- msmsgs.exe PID 548 0x520000 vad_tag Vad PAGE_EXECUTE_READ (RX, not RWX)
- plugin volatility3.windows.malfind.Malfind

SOURCE ARTIFACT /cases/AMF_MemorySamples/windows/sample001.bin · SHA-256 03242077eb3364fb...316d561c

SEAL hmac-sha256:3788bca5030d85513b3d09b9c761d6f44642635f222049ac544979d8f331985f · provenance MCP · run 4ecbae25

LOW F-AMF-S001-006

Process inventory recovered: 21 running processes, coherent PPID forest (2 roots, 0 orphans, 0 LOLBin flags)

get_pslist returned 21 processes (System pid4, smss 356, csrss 604, winlogon 628, services 680, lsass 692, svchost 852...); build_process_tree: process_count 21, root_count 2 (System, smss.exe), orphan_count 0, suspicious_count 0. cmdline recovered all 21 command lines consistent with pslist. get_netscan recovered 0 sockets (honest empty real result on this XP-era image). Baseline process-discovery context for the injection findings.

MITRE T1057 · confidence 0.6 · host sample001

EVIDENCE REFS

- get_pslist process_count=21
- build_process_tree: process_count 21, root_count 2 (System pid4, smss.exe pid356), orphan_count 0, suspicious_count 0
- get_netscan socket_count=0
- get_svcscan service_count=229
- cmdline rows=21

SOURCE ARTIFACT /cases/AMF_MemorySamples/windows/sample001.bin · SHA-256 03242077eb3364fb...316d561c

SEAL hmac-sha256:39c2d1ea7c967e4cc891333b85e0d5dbbd8a804be2e5372e0b9f6063d9706aba · provenance MCP · run 82ad0e4d

Timeline (UTC)

TIME (UTC)	HOST	EVENT	TECHNIQUE	SOURCE
2026-06-06T20:36:37Z	sample001	Case opened case_init: AMF-WIN-SAMPLE001 created and set active, examiner victor.galvan, severity medium, scope /cases/AMF_MemorySamples/windows/sample001.bin	—	case_init
2026-06-06T20:48:49Z	sample001	Evidence registered (SHA-256 custody anchor) evidence_register: sample001.bin sha256 03242077eb3364fb	—	evidence_register

TIME (UTC)	HOST	EVENT	TECHNIQUE	SOURCE
		248d1c7730fd0a94 074583df8deb2606 d6c93c31316d561c , size 536330240 bytes, indexed to agentropix- evidence- 2026.06.06		
2026-06-06T20:50:00Z	sample001	Memory triage: malfind recovered 15 executable VAD hits (11 RWX / 4 RX) get_malfind (volatility3.win dows.malfind.Mal find) completed in 75s: 15 executable VAD hits - 11 PAGE_EXECUTE_REA DWRITE (RWX) + 4 PAGE_EXECUTE_REA D (RX). RWX by process: winlogon.exe x9 (PID 628), csrss.exe x1 (PID 604), msimn.exe x1 (PID 1984). RX (not RWX): winlogon.exe x1 (PID 628, 0x580000), lsass.exe x2 (PID 692), msmsgs.exe x1 (PID 548).	T1055	get_malfind
2026-06-06T20:50:05Z	sample001	Memory triage: process/service/net work/tree enumeration get_pslist 21 processes; get_netscan 0 sockets (empty real result, XP- era image); get_svcscan 229 services; build_process_tr ee clean (2 roots, 0 orphans, 0 suspicious)	T1057	get_pslist/get_n etscan/get_svcsc an/build_process _tree
2026-06-06T23:17:43Z	sample001	Finding F-AMF- S001-001 approved (SIMULATED)	-	approve_finding

TIME (UTC)	HOST	EVENT	TECHNIQUE	SOURCE
		examiner approval, demo only) approve_finding DRAFT→APPROVED for F-AMF-S001- 001, examiner victor.galvan, approval_id 4a881577139b59ef adb980816d47adfc ecbda4ad6bb94fd9 2fa8a797973696b4 . NOTE: Playwright- automated demo approval, not a human HMAC sign- off.		

Case opened 2026-06-06 20:36 UTC and the 511 MiB image was registered with its SHA-256 custody anchor at 20:48. Memory triage (pslist / netscan / svcscan / process-tree) ran at 20:50, with malfind completing its 75 s pass over the same image. Finding F-AMF-S001-001 was approved at 23:17 (Playwright-automated demo approval — not a human HMAC sign-off). The host's own RAM clock shows a boot at 2012-11-26 22:03:28 UTC and the memory acquisition tool (mdd.exe) starting 2012-11-27 01:57:28 UTC.

Agentropix Performance

The full triage processed a 536,330,240-byte (511 MiB) raw image. The heaviest stage, malfind, exceeded the 180 s SDK request bound on the first pass and was re-run standalone with a 300 s callTool timeout, completing in 75 s. SHA-256 hashing of the image is the chain-of-custody anchor. Recall figures are the canonical Agentropix corpus benchmarks (facts.md), not per-case measurements.

~75 s

TOTAL RUNTIME
malfind heaviest stage

11

MCP TOOL CALLS
health, case_init,
case_status,
evidence_register,
get_image_info, pslist,
netscan, malfind, svcscan,
build_process_tree,
cmdline

511 MiB

EVIDENCE PROCESSED
536,330,240 bytes raw

100% /
91.5%

RECALL

disk 72/72 · memory
108/118 (canonical)

STAGE / TOOL	CAPTURE	DURATION	RESULT
evidence_register (SHA-256)	sample001.bin	—	536,330,240 B · sha256 0324...d561c
get_pslist	processes	<5 s	21 processes
get_netscan	sockets	<5 s	0 sockets (empty real result)
get_malfind	RWX/RX VADs	75 s	15 hits (11 RWX / 4 RX) — re-run @300 s timeout
get_svcscan	services	<10 s	229 services (118 running)
build_process_tree	PPID forest	<5 s	2 roots, 0 orphans, 0 suspicious
run_volatility cmdline	command lines	<10 s	21 rows
report_generate	profile=full	<5 s	6 approved findings, sealed

Coverage Attestation

This case is a single memory image, so corpus-vs-parsed cross-check is not applicable per-case; the recall figures below are the canonical Agentropix benchmark suite results (facts.md). The EvidenceGate was respected throughout: only APPROVED findings appear in this sealed report, and the non-dry-run finding write required a single-scope evidence-gate mutation token.

Disk recall

72/72 (100%)

Memory recall

108/118 (91.5%)

Tests passing

4464

Approval gate

APPROVED-only

This report is sealed (report_id 22560e04e9b123a0...) and contains 6 APPROVED findings only. Every datum is grounded in the case's real MCP captures (get_malfind / get_pslist / get_svcscan / build_process_tree / get_netscan) and the SHA-256 custody anchor. The F-AMF-S001-001 approval was Playwright-automated for the demo and is NOT a human HMAC sign-off — a production case requires the examiner's portal sign-off.

Recommendations

P1 Dump and statically analyse the 11 RWX VAD regions — especially the 9 in `winlogon.exe` (PID 628, base addresses 0x42e20000, 0x22f40000, 0x24c00000, 0x248a0000, 0x40a60000, 0x2b7f0000, 0x43400000, 0x57500000, 0x548a0000) — with `vol3 vadinfo/dlldump` to confirm whether the injected code is malicious or benign packing.

P2 Obtain human examiner HMAC sign-off in the Examiner Portal for the six findings — the F-AMF-S001-001 approval in this run was Playwright-automated (demo only) and must be re-attested before any courtroom use.

P2 Correlate the in-image acquisition tooling (`mdd.exe` PID 244, `wc.exe` PID 364) with the documented acquisition procedure to confirm the RAM capture chain-of-custody and rule out acquisition-induced artefacts.

P3 Treat RWX in GUI apps (`msimn.exe` Outlook Express, `msmsgs.exe` Messenger) as low-confidence — on XP-era images these are commonly benign JIT/packing; deprioritise unless static analysis says otherwise.

P3 No network IOCs are recoverable from RAM; if a disk image or pcap for this host exists, ingest it to anchor an initial-access vector and C2 channel that the memory-only scope cannot supply.

Appendix

Methodology

Generated by the Agentropix-SIFT report engine (ADR-024): the MCP server projects the case's sealed findings/timeline/IOC indices into a full-profile snapshot, EvidenceGate-filtered to APPROVED documents, then HMAC-seals the roll-up. The raw forensic captures came from the \$3.A manual triage sequence run live against /cases/AMF_MemorySamples/windows/sample001.bin on 2026-06-06 via the Agentropix MCP at <TAILNET-HOST>:8765/mcp (get_pslist, get_netscan, get_malfind, get_svcscan, build_process_tree, run_volatility cmdline). No figure in this report is simulated.

Tool & framework versions

Python

3.12+

Volatility 3

windows.malfind.Malfind · windows.netscan.NetScan · windows.cmdline.CmdLine plugins

Agentropix-SIFT MCP

server 3.2.4 · 72 tools (71 forensic + health)

mermaid-cli

@mermaid-js/mermaid-cli (inlined mermaid.min.js)

Report engine

ADR-024 · profile=full · HMAC-sha256 seal

Test suite

4464 tests passing

Chain-of-custody hashes

ARTEFACT	ALGO	DIGEST	STATUS
sample001.bin (evidence image)	SHA-256	03242077eb3364fb248d1c7730fd0a94074583df8deb2606d6c93c31316d561c	verified custody anchor
evidence record id	SHA-256	f2649687665ca02992131101006024b1cf994aa8b8270864fda216687e92bd7d	indexed
report (full profile)	report_id	22560e04e9b123a0d3e09776cb5cc4cca2282c1108e009e03ed12e691612a33a	sealed
F-AMF-S001-001 seal	HMAC-sha256	hmac-sha256:fd05a9d4c2721e5a693b03b55d06ceea7cb e8429056fa93338800bae b8ed9cee	approved (demo)

Provenance: every claim traces to a real MCP tool output or the SHA-256 custody anchor recorded in the case EXECUTED-RUN. The image is the Art-of-Memory-Forensics teaching sample; classification is

RESTRICTED·FORENSIC. MCP host shown as <TAILNET-HOST> — no raw internal hostnames or secrets appear in this document.

SEALED — report_id 22560e04e9b123a0..., 6 approved findings (HMAC-sha256) · Generated by Agentropix-SIFT report engine (ADR-024) · MCP host · 2026-06-07T04:21:03Z

All data sourced from sealed report_generate(profile=full) sections and the case EXECUTED-RUN captures. No facts fabricated.